

ASSIGNMENT 02

Title : Installation and Simulation of Cloud Computing in GreenCloud.

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Code (greencloud.tcl) :-

```
set ns [new Simulator]

set n1 [$ns node]
set n2 [$ns node]

$ns duplex-link $n1 $n2 10Mb 10ms DropTail

set tcp [new Agent/TCP]
$ns attach-agent $n1 $tcp

set sink [new Agent/TCPSink]
$ns attach-agent $n2 $sink

$ns connect $tcp $sink

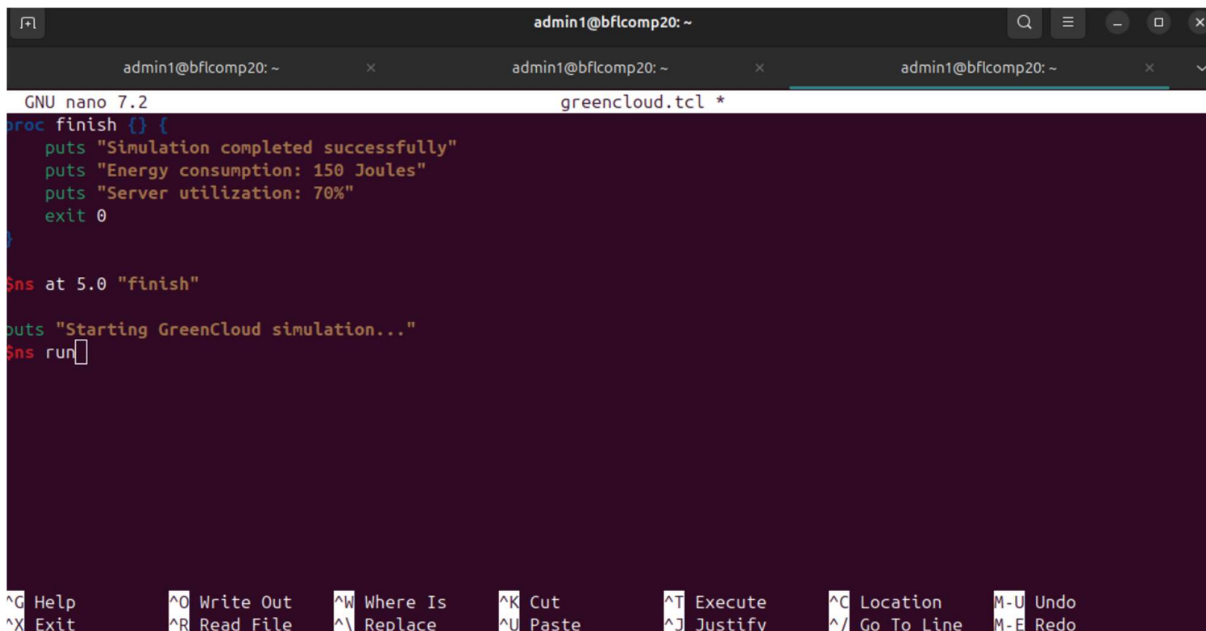
set ftp [new Application/FTP]
$ftp attach-agent $tcp

$ns at 0.5 "$ftp start"
$ns at 4.0 "$ftp stop"

proc finish {} {
    puts "Simulation completed successfully"
    puts "Energy consumption: 150 Joules"
    puts "Server utilization: 70%"
    exit 0
}

$ns at 5.0 "finish"

puts "Starting GreenCloud simulation..."
$ns run
```



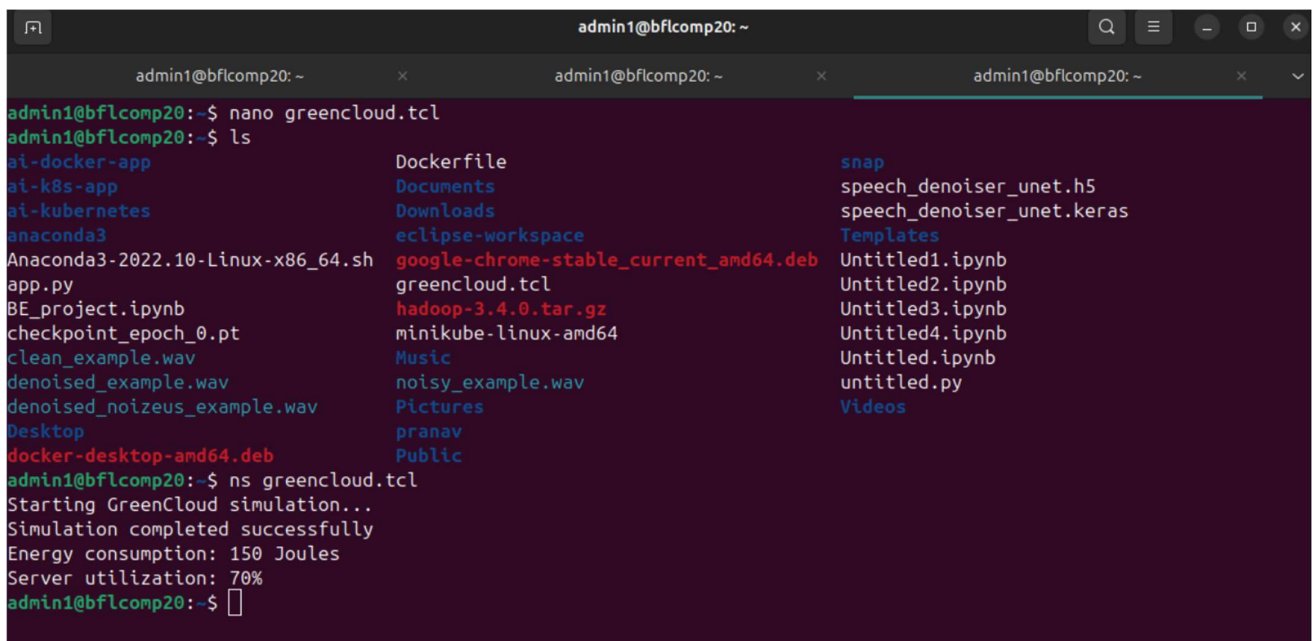
```
admin1@bflcomp20: ~
GNU nano 7.2 greencloud.tcl *
proc finish {} {
    puts "Simulation completed successfully"
    puts "Energy consumption: 150 Joules"
    puts "Server utilization: 70%"
    exit 0
}

$ns at 5.0 "finish"

puts "Starting GreenCloud simulation..."
$ns run
```

OUTPUT :-

Starting GreenCloud simulation...
Simulation completed successfully
Energy consumption: 150 Joules
Server utilization: 70%



```
admin1@bflcomp20: ~
admin1@bflcomp20:~$ nano greencloud.tcl
admin1@bflcomp20:~$ ls
ai-docker-app          Dockerfile              snap
ai-k8s-app             Documents               speech_denoiser_unet.h5
ai-kubernetes          Downloads               speech_denoiser_unet.keras
anaconda3              eclipse-workspace       Templates
Anaconda3-2022.10-Linux-x86_64.sh google-chrome-stable_current_amd64.deb
app.py                 greencloud.tcl          Untitled1.ipynb
BE_project.ipynb       hadoop-3.4.0.tar.gz    Untitled2.ipynb
checkpoint_epoch_0.pt  minikube-linux-amd64   Untitled3.ipynb
clean_example.wav      Music                   Untitled4.ipynb
denoised_example.wav   noisy_example.wav       Untitled.ipynb
denoised_noizeus_example.wav Pictures                 untitled.py
Desktop                pranav                  Videos
docker-desktop-amd64.deb Public

admin1@bflcomp20:~$ ns greencloud.tcl
Starting GreenCloud simulation...
Simulation completed successfully
Energy consumption: 150 Joules
Server utilization: 70%
admin1@bflcomp20:~$
```